

AI Soccer Arena — Game Specification

1. Game Overview

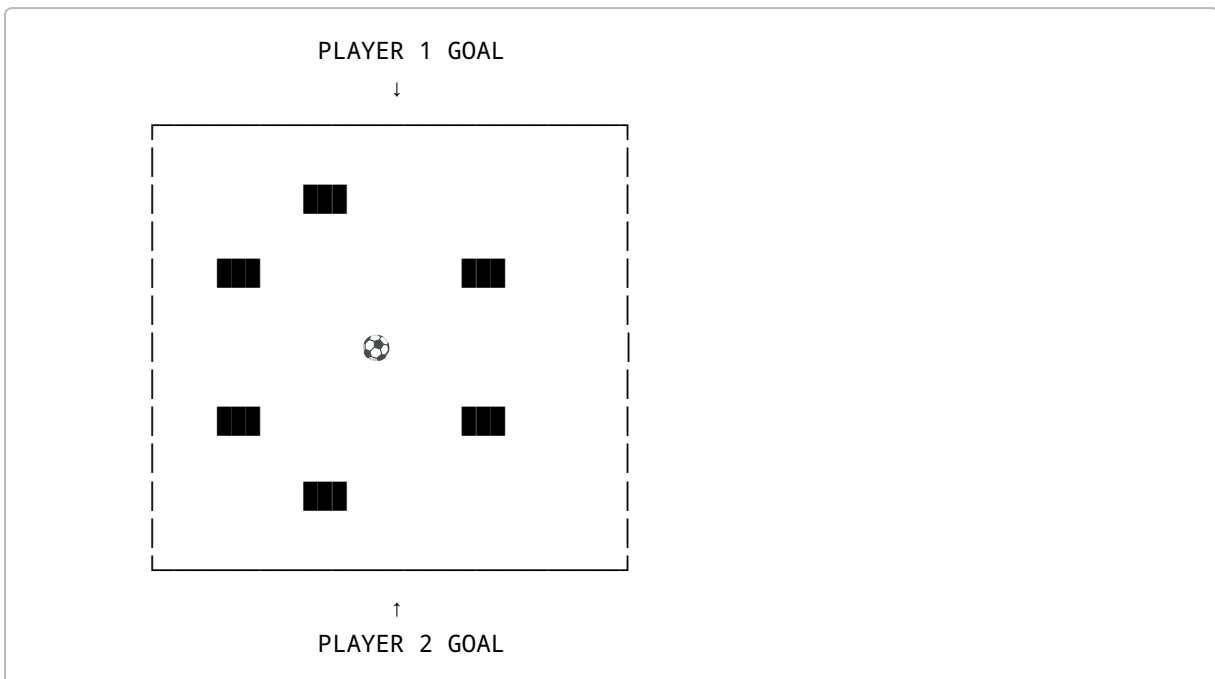
The game is a 2D soccer simulation played between two AI-controlled players and one ball. The field is a flat rectangular arena containing two goals placed on opposite sides and a fixed number of procedurally generated obstacles.

Both players have identical movement and interaction capabilities. There are no permanent attacker or defender roles. Both players can freely move around the entire field, pursue the ball, defend their goal, and attack the opponent's goal.

The objective is to score more goals than the opposing player before the game ends.

2. Field

The playing area is a rectangular 2D field with fixed dimensions. The two goals are located on opposite sides of the field.



The field is completely flat. There are no slopes, elevation changes, or terrain effects.

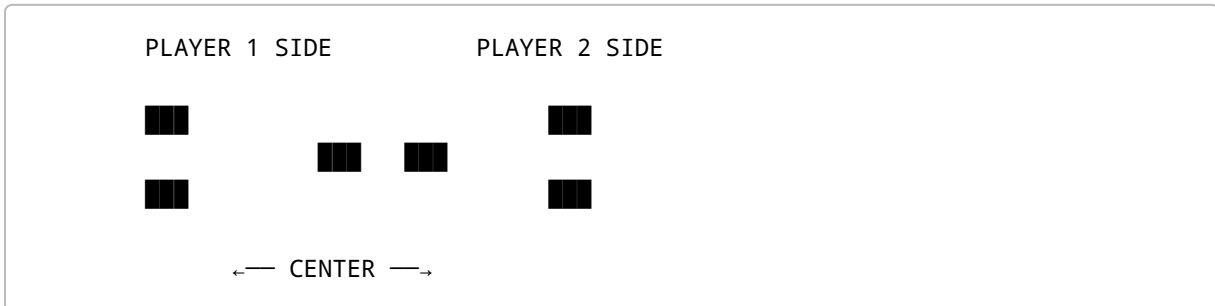
The outer boundary of the field acts as a solid wall. Neither player nor the ball can leave the playable area.

3. Obstacles

Each match contains a fixed number of static obstacles.

Obstacle positions are generated using a random seed at the beginning of the match. The obstacle layout is mirrored across the center of the field so that both sides have equivalent geometry.

For example:



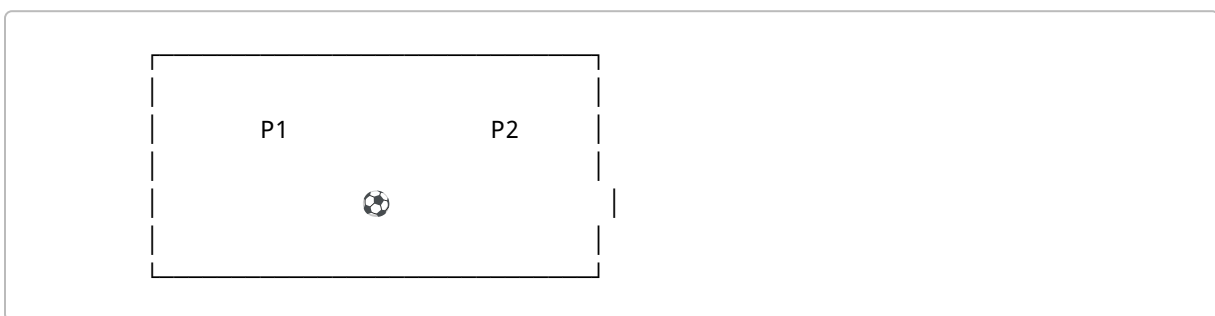
Obstacles cannot overlap the goals, boundary walls, player starting positions, or initial ball position.

Obstacles remain stationary throughout the match.

Different random seeds produce different obstacle configurations.

4. Players

There are exactly two players, controlled independently.



Both players have identical movement speed, size, collision properties, and kicking capabilities.

Players can move freely throughout the field and are not restricted to specific areas.

Players cannot move outside the field or pass through obstacles.

There are no goalkeepers, fixed positions, teams, fouls, offside rules, or other conventional soccer rules.

5. Game Iterations

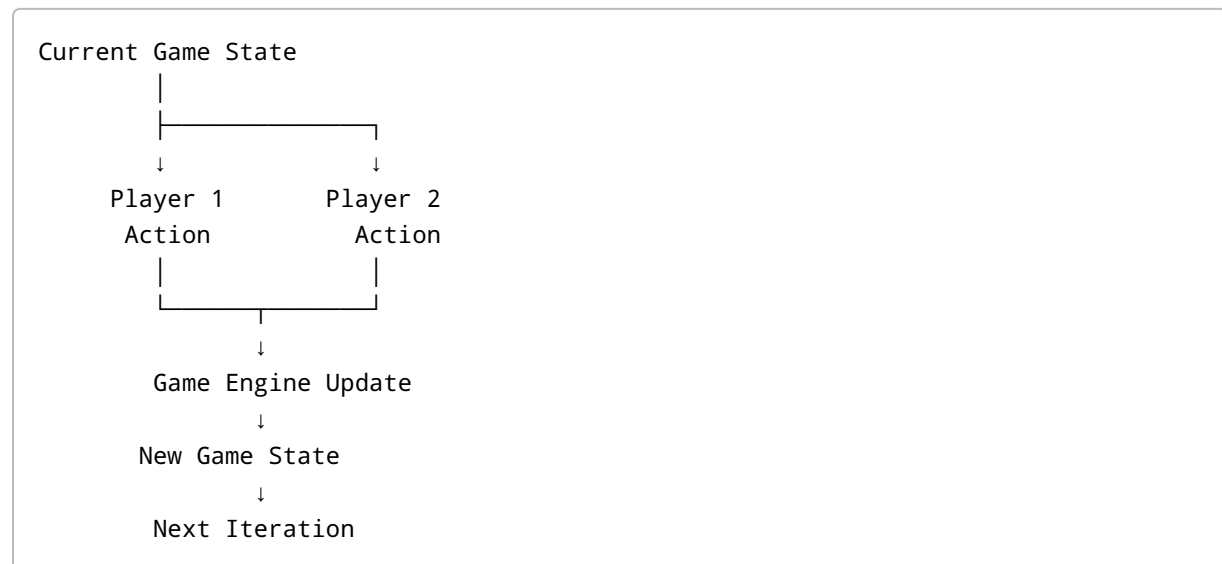
The game operates in discrete iterations.

At the beginning of every iteration, both players receive the current game state and independently select their actions.

Both actions are applied simultaneously by the game engine.

The environment then updates player positions, ball movement, collisions, possession, and goal conditions before producing the next game state.

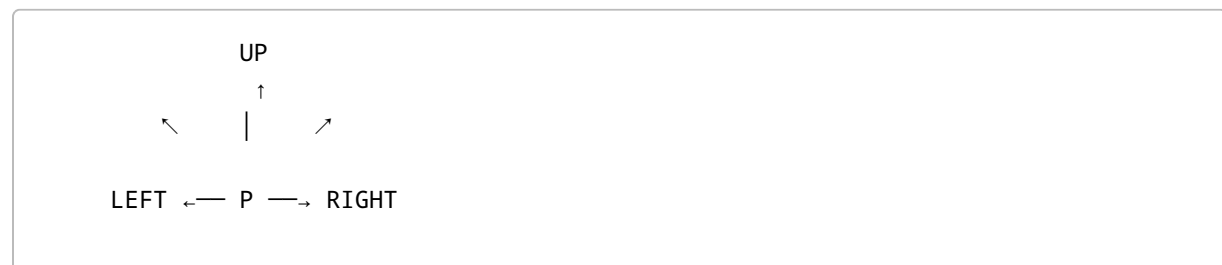
The game therefore follows:



6. Player Movement

Each player chooses a movement direction during every iteration.

The initial version supports eight movement directions and an optional stationary action:





Players move at a fixed speed. The speed is identical for both players and does not change during the match.

Movement is simultaneous for both players.

If a player's intended movement would cause them to enter an obstacle or leave the field, the game engine resolves the movement according to the predefined collision rules.

7. Ball

There is exactly one ball in the game.

P1



P2

The ball can be stationary, possessed by a player, or moving through the field after a kick.

The ball has a fixed size and follows deterministic movement and collision rules.

8. Possession

Possession is determined automatically by the game engine.

When a player reaches the ball within the defined possession radius, that player obtains possession.

Players do not explicitly choose a "possess" action.

Only one player can possess the ball at a time.

If both players reach the ball during the same iteration, a fixed deterministic tie-breaking rule is used.

9. Kicking

A player can kick the ball only while in possession.

A kick determines the direction and travel distance of the ball.

The player chooses the kick direction and the allowed kick distance/power according to the predefined action limits.

Once kicked, the ball becomes independent of the player and follows its trajectory through the field.



The player does not retain control of the ball while it is travelling.

10. Ball Movement

The ball moves at a predefined speed according to the direction selected for the kick.

Each kick has a predefined maximum travel distance.

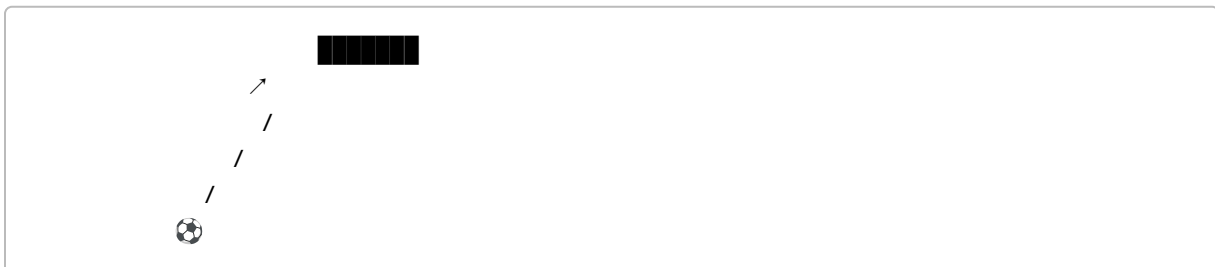
The ball continues moving across subsequent game iterations while both players continue making movement decisions.

A kick ends when its allowed travel distance is exhausted or when a terminating event such as a player collision or goal occurs.

11. Wall and Obstacle Bounces

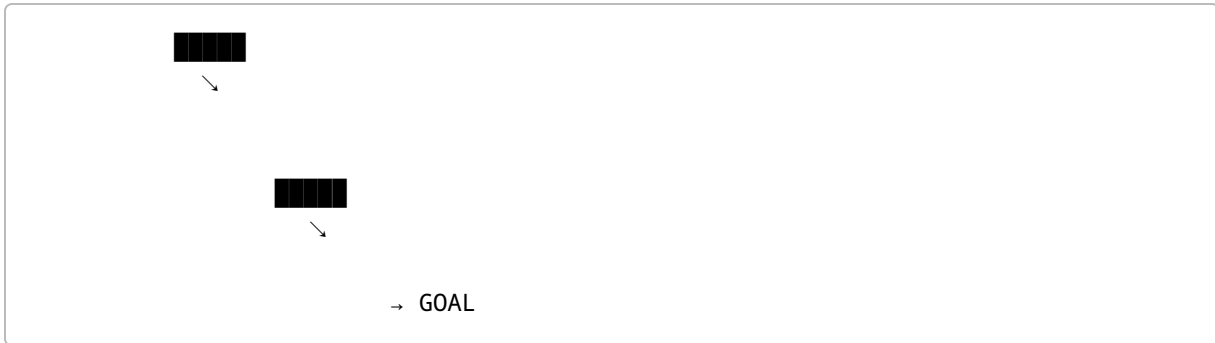
The outer field boundaries and obstacles behave as solid surfaces.

When the ball hits a wall or obstacle, it reflects from the surface.



The direction of the ball changes according to the angle of impact, while the remaining travel distance of the current kick is preserved.

Therefore, a single kick can produce multiple rebounds.

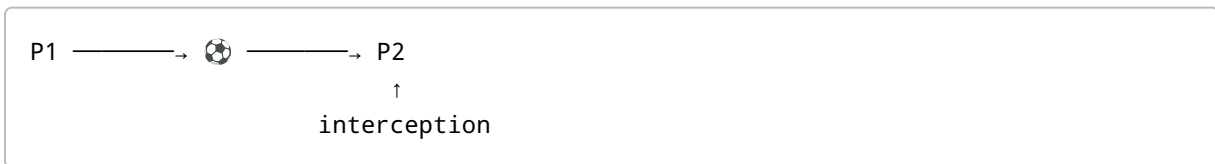


This allows indirect or bank shots around obstacles.

12. Ball-Player Collision

A moving ball can collide with either player.

If the ball reaches the opposing player while travelling, the current ball trajectory stops and the opposing player obtains possession.



This allows players to intercept an opponent's kick while the ball is moving.

The collision behavior is identical and deterministic for both players.

13. Player-Player Collision

Players cannot occupy the same physical space.

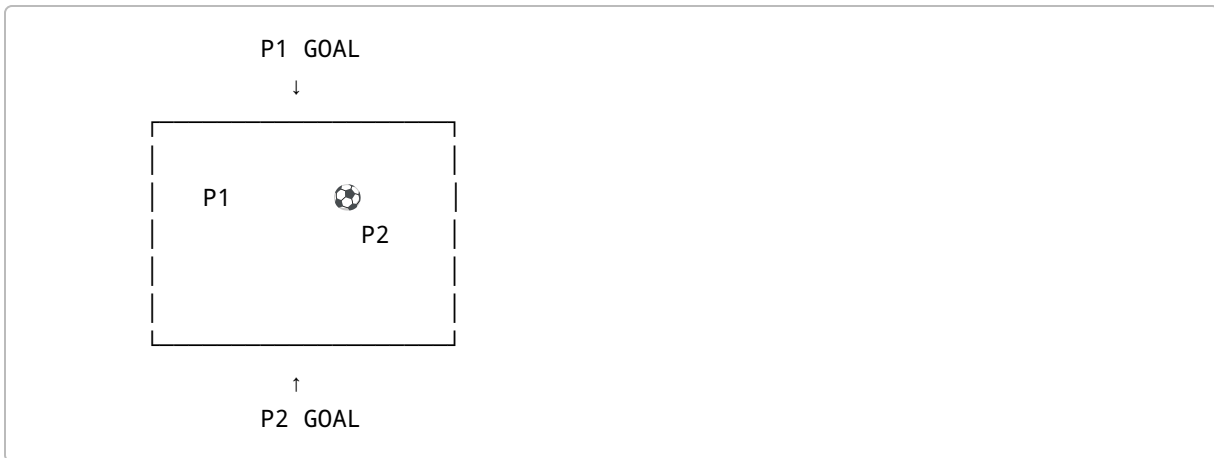
If the two players collide, the environment resolves the collision using a fixed deterministic rule.

A player-player collision does not directly result in a goal, foul, penalty, or automatic change of possession.

14. Goals

There are exactly two goals, positioned on opposite sides of the field.

Each player attempts to score in the opponent's goal while defending their own.



A goal is registered when the ball enters the corresponding goal area according to the predefined goal-entry condition.

Once a goal is registered, the current play immediately ends and the game enters the restart phase.

15. Goal Restart

After every goal, the field is reset to the standard starting configuration.

Both players are returned to their predefined restart positions and the ball is placed at the predefined kickoff position.

The player who conceded the previous goal receives possession.

For example:

```
Player 1 scores
      ↓
Player 2 receives kickoff
```

After the next goal:

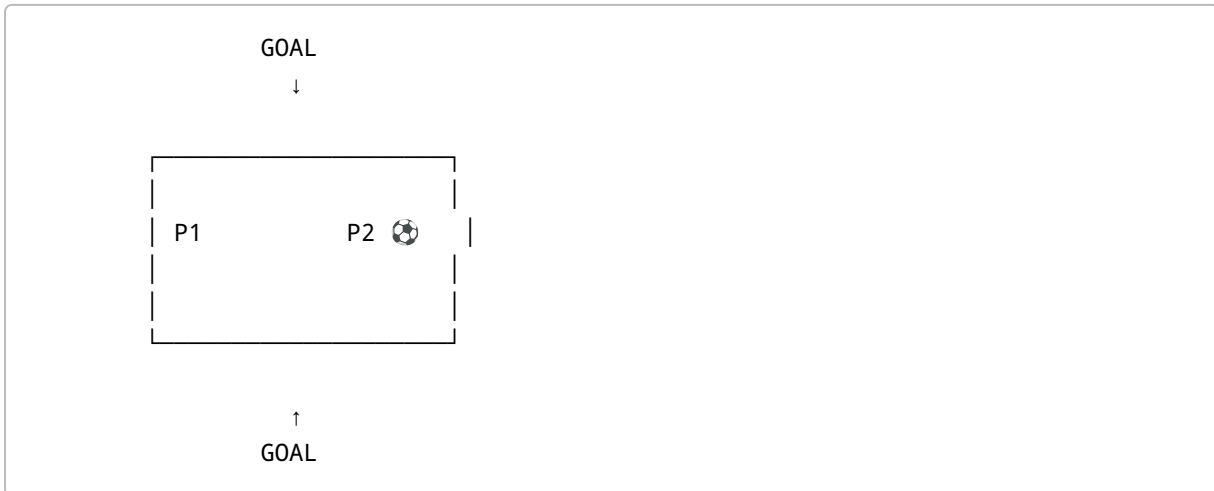
```
Player 2 scores
      ↓
Player 1 receives kickoff
```

This is only a restart mechanism. It does not create attacker or defender roles.

Once the restart occurs, both players are again free to move anywhere on the field.

16. Starting Configuration

Every match begins from a standardized starting configuration.



The starting configuration is symmetric, with one player initially possessing the ball.

Initial possession can be assigned according to a fixed rule, and subsequent possession alternates after every goal.

17. Game Duration

Every match has a fixed maximum number of game iterations.

The match also has a predefined maximum number of goals.

The game ends when either the maximum iteration limit or maximum goal limit is reached.

If the iteration limit is reached before the maximum goal count, the player with more goals wins.

If both players have the same score when the game ends, the match is recorded as a draw unless a competition-level tiebreaker is defined.

18. Randomness and Reproducibility

Every match is associated with a random seed.

The seed determines the procedural obstacle configuration and any other randomized environmental elements.

The same seed, starting state, and sequence of player actions must always produce the same game outcome.

This allows matches to be reproduced, tested, replayed, and independently verified.

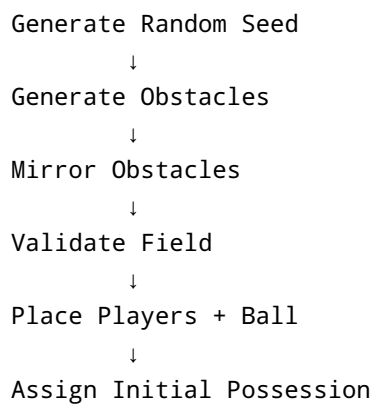
19. Fixed Game Parameters

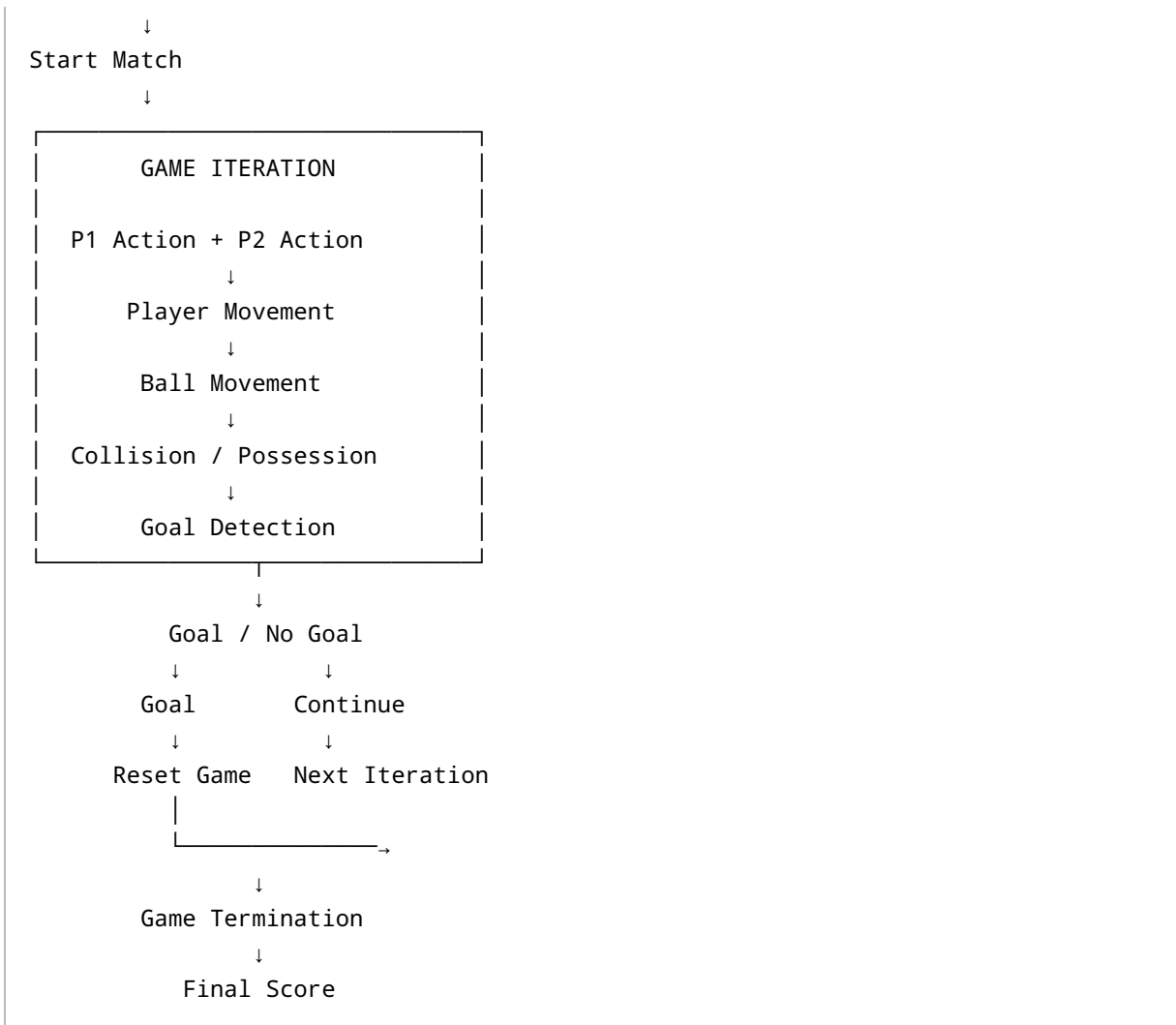
The following parameters are fixed for a competition round:

- Field dimensions
- Goal dimensions and positions
- Player size
- Player movement speed
- Ball size
- Ball speed
- Kick distance/range
- Possession radius
- Number of obstacles
- Obstacle dimensions
- Maximum game iterations
- Maximum goals
- Collision behavior
- Starting positions
- Restart positions

All participants are evaluated using the same environment configuration.

20. Match Flow





The game engine is responsible for enforcing all game rules, movement restrictions, collision handling, ball physics, possession, goal detection, and match termination.

The AI players only provide their actions at each game iteration.